Near-infrared spatiotemporal color vision in humans enabled by upconversion contact lenses

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Abstract

Humans cannot perceive infrared light due to the physical thermodynamic properties of photon-detecting opsins. However, the capability to detect invisible multispectral infrared light with the naked eye is highly desirable. Here, we report wearable near-infrared (NIR) upconversion contact lenses (UCLs) with suitable optical properties, hydrophilicity, flexibility, and biocompatibility. Mice with UCLs could recognize NIR temporal and spatial information and make behavioral decisions. Furthermore, human participants wearing UCLs could discriminate NIR information, including temporal coding and spatial images. Notably, we have developed trichromatic UCLs (tUCLs), allowing humans to distinguish multiple spectra of NIR light, which can function as three primary colors, thereby achieving human NIR spatiotemporal color vision. Our research opens up the potential of wearable polymeric materials for non-invasive NIR vision, assisting humans in perceiving and transmitting temporal, spatial, and color dimensions of NIR light.

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